

Homework 7

For Thursday, October 8, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. You may use the results covered through Lecture 8, including the chain rule, the inverse-function derivative theorem, and the derivatives of $\ln$ and $\arcsin$. For implicit curves, you may assume that the local branches used are differentiable wherever the differentiated equation has a nonzero coefficient of the derivative being solved for.

1. Write $F(x) = (2 + \cos(3x))^4$ as a composition of three functions. Use the chain rule to find $F'(x)$.
2. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Suppose

$$\begin{array}{llll} f(1) = 3, & g(1) = 2, & & \\ f'(1) = 2, & g'(1) = -1, & f'(2) = 4, & g'(3) = 5. \end{array}$$

Find $(f \circ g)'(1)$ and $(g \circ f)'(1)$.

3. Let

$$F(x) = \frac{e^{2x}}{1 + e^{2x}}, \quad x \in \mathbb{R}.$$

Find and simplify $F'(x)$. Determine every real x for which $F'(x) = 1/2$.

4. Let $F(x) = \ln(1 - x^2)$. Find its domain, compute its derivative, and state the domain of the derivative.
5. Suppose f is differentiable on an open interval containing 2. The graph $y = f(t)$ has tangent line

$$y = 5 - 3(t - 2)$$

at $t = 2$. Find the tangent line to $y = f(x^2 + 1)$ at $x = 1$.

6. Evaluate the limit by interpreting it as a derivative:

$$\lim_{h \rightarrow 0} \frac{\ln(1 + 3h)}{h}.$$

7. Determine all constants $a, b \in \mathbb{R}$ for which

$$F(x) = \begin{cases} ae^{2x} + b, & x < 0, \\ \ln(1 + x), & x \geq 0 \end{cases}$$

is differentiable at zero. Find $F'(0)$ for those constants.

8. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable and $f(f(x)) = x$ for every real x . Prove that $f'(x) \neq 0$ for every x . If $f(a) = a$, what are the possible values of $f'(a)$?

9. For the curve

$$e^y + xy = e^x,$$

find dy/dx where it can be obtained by solving the differentiated equation. Check that $(0, 0)$ lies on the curve and find its tangent line there.

10. Find all points of the curve

$$x^2 + xy + y^2 = 3$$

where the tangent is horizontal. Give the tangent lines and verify that the denominator in your formula for dy/dx is nonzero at each point found.

11. Consider the curve $x = y^2$. Regard x as a function of y and find the tangent line at $(0, 0)$. Explain why neither branch $y = \sqrt{x}$ nor $y = -\sqrt{x}$, defined for $x \geq 0$, has a finite right derivative at zero.
12. The curve $y^2 = x^4$ is the union of two differentiable graphs $y = y(x)$. Find these two graphs and their tangent lines at the origin. What does implicit differentiation give at the origin, and does that equation determine the tangent lines?
13. The function $f(t) = t^3 + 2t$ is one-to-one from $\mathbb{R}$ onto $\mathbb{R}$. Let $g = f^{-1}$. Find $g'(3)$ without finding a formula for g .

14. The function

$$\tan : (-\pi/2, \pi/2) \longrightarrow \mathbb{R}$$

is one-to-one and onto, with inverse $\arctan$. Use the inverse-function derivative theorem to find $\frac{d}{dx} \arctan x$, expressing your answer entirely in terms of x . Explain why the derivative exists for every real x .

15. Let $f(t) = t^3$ and $g = f^{-1}$, so $g(x) = \sqrt[3]{x}$. Use the inverse-function derivative theorem to find $g'(x)$ for $x \neq 0$. Then use the definition of the derivative to decide whether $g'(0)$ exists. Which hypothesis of the theorem fails at zero?

16. Let $F(x) = \arcsin(\sin x)$ for $-\pi \leq x \leq \pi$. Write F as a piecewise formula without trigonometric functions. Find $F'(x)$ wherever it exists in $(-\pi, \pi)$, and explain what happens at the remaining interior points.

17. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$f(x+y) = f(x)f(y) \quad \text{for all } x, y \in \mathbb{R}, \quad f(0) = 1,$$

and is differentiable at zero. Use the definition of the derivative to prove that f is differentiable at every $a \in \mathbb{R}$ and that

$$f'(a) = f(a)f'(0).$$

18. Define

$$g(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

- (a) Prove from the definition that $g'(0) = 0$.
- (b) Use the chain rule to find $H'(0)$ for $H(x) = e^{g(x)}$.
- (c) Give infinitely many nonzero h , arbitrarily close to zero, for which $g(h) = g(0)$. Explain why a proof of part (b) cannot divide by $g(h) - g(0)$ for every sufficiently small nonzero h .
19. Find nonconstant functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that f is differentiable everywhere, g is continuous everywhere but not differentiable at zero, and $f \circ g$ is differentiable at zero. Justify all these properties and find $(f \circ g)'(0)$.
20. The graph of f consists of the two straight rays shown; each ray continues indefinitely. Let $H(x) = f(x^2)$. Find $H'(x)$ wherever it exists, and determine all points where H is not differentiable.

